

An improved recursive decomposition algorithm for reliability evaluation of lifeline networks

Liu Wei^{1,2†} and Li Jie^{1,2‡}

1. *Department of Building Engineering, Tongji University, Shanghai 200092, China*

2. *State Key Laboratory for Disaster Reduction in Civil Engineering, Tongji University, Shanghai 200092, China*

Abstract: The seismic reliability evaluation of lifeline networks has received considerable attention and been widely studied. In this paper, on the basis of an original recursive decomposition algorithm, an improved analytical approach to evaluate the seismic reliability of large lifeline systems is presented. The proposed algorithm takes the shortest path from the source to the sink of a network as decomposition policy. Using the Boolean laws of set operation and the probabilistic operation principal, a recursive decomposition process is constructed in which the disjoint minimal path set and the disjoint minimal cut set are simultaneously enumerated. As the result, a probabilistic inequality can be used to provide results that satisfy a prescribed error bound. During the decomposition process, different from the original recursive decomposition algorithm which only removes edges to simplify the network, the proposed algorithm simplifies the network by merging nodes into sources and removing edges. As a result, the proposed algorithm can obtain simpler networks. Moreover, for a network owning s -independent components in its component set, two network reduction techniques are introduced to speed up the proposed algorithm. A series of case studies, including an actual water distribution network and a large urban gas system, are calculated using the proposed algorithm. The results indicate that the proposed algorithm provides a useful probabilistic analysis method for the seismic reliability evaluation of lifeline networks.

Keywords: lifeline system; network reliability; path-based recursive decomposition algorithm; disjoint minimal path; disjoint minimal cut; network reduction; reliability bound

1 Introduction

A main challenge in the reliability analysis of lifeline systems, such as water distribution and urban gas supply networks, is how to evaluate the node-pair reliabilities (Li, 2005). The methods proposed by in previous research can be divided into two major categories: the Monte Carlo simulation approach and the probabilistic analytical algorithm. For the Monte-Carlo simulation approach, the precision of the simulation results cannot be easily estimated (Li, 2005). Therefore, many efforts have been focused on exploring the probabilistic analytical algorithm, which can theoretically obtain accurate results.

Since network reliability can be calculated by adding the probability of all disjoint minimal paths, path-based algorithms have been widely investigated by many researchers. In 1973, Fratta and Montanari (1973) first adopted the sum of disjoint products (SDP) method

to compute network reliability. In 1975, Aggarawal and Misra (1975) introduced Boolean algebra into the SDP method, which simplified the disjoint process of minimal paths. Then, Abranham (1979) and Liao (1982a and b), presenting the so called "four sum of disjoint theorems" respectively, further developed the disjoint minimal path method and developed the path-based algorithms into a classical method. Later, in order to improve efficiency and apply the method to complex networks, and combine it with series-edge and parallel-edge reduction technologies, Page and Perry (1988; 1989) adopted the factoring theorem to reduce the complexity of network reliability analysis. However, the method cannot simplify a network into a simplest network whose reliability can be calculated directly. Thus, its application is limited. A common characteristic of the above methods is that all disjoint products have to be enumerated to provide an accurate result. However, as the problem to search for all network paths is a non-polynomial hard (NP hard) problem, i.e., the number of the paths in a network can be expressed as an exponential function instead of a polynomial function of the number of the network edges or nodes, the above approaches are not available for use with large lifeline networks. For large lifeline networks, an approximate algorithm should be developed to evaluate the reliability which can satisfy a prescribed error bound. Dotson

Correspondence to: Li Jie, Department of Building Engineering, Tongji University, Shanghai 200092, China
Tel: +86-21-65981505, Fax: +86-21-65986345
E-mail: lijie@mail.tongji.edu.cn

[†]Lecturer; [‡]Professor

Supported by: Natural Science Funds for the Innovative Research Group of China Under Grant No. 50621062

Received December 10, 2008; **Accepted** March 9, 2009

and Gobien (1979) first proposed a real-time disjoint of minimal paths in a network and gave an example to calculate the approximate value of network reliability. Later in 1988, by replacing the edge-incidence matrix with an adjacency matrix and introducing the breadth-first search technology, a modified Dotson algorithm was developed by Yoo and Deo (1988). Also, Yoo and Deo (1988) compared four current algorithms and concluded that Dotson's algorithm was the most efficient. Introducing the concept of "system structure function" and combining Dotson's algorithm with computer storage, Li and He (2002) presented a recursive decomposition algorithm, in which the probabilistic inequality is used to control the computation time and ensure the error bound remained within the predefined scope. This algorithm provides an efficient approach for large network reliability evaluation. Despite the detailed descriptive algorithm, rigorous mathematical proof of the method has yet to be achieved.

In this paper, using the concept of system structure function and minimal path set, three theorems are introduced and proved. Based on these theorems and the original recursive decomposition algorithm, an improved algorithm is derived to calculate the network reliability. The proposed algorithm takes the shortest path from the source to the sink of a network as decomposition policy. Using the Boolean laws of set operation and the probabilistic operation principal, a recursive decomposition process is constructed. During the decomposition process, different from the original recursive decomposition algorithm which only removes edges to simplify the network, the proposed algorithm simplifies the network by merging nodes into sources and removing edges. As a result, the proposed algorithm can obtain simpler networks. Also, when decomposing the network, the disjoint minimal path set and the disjoint minimal cut set are simultaneously enumerated. Consequently, the probabilistic inequality can be used to yield results that satisfy a prescribed error bound. Moreover, for a network owning s -independent components in its component set, two network reduction techniques are introduced to speed up the proposed algorithm. The proposed algorithm has been applied to a series of case studies. The results show that it provides a useful probabilistic analysis tool for the seismic reliability evaluation of lifeline networks.

2 Structure function and related theorems

A lifeline network is a graph with a weight, defined as the success probability (reliability) subject to seismic wave excitation, assigned to each edge or node. In general, network-assigned weight can be classified into three categories: (1) edge weighted network, where only edges are assigned weights; (2) node weighted network, where only nodes are assigned weights; and (3) general weighted network, where both nodes and edges are assigned weights. In this paper, only an edge weighted

network is considered. However, for the node weighted network and general weighted network, the proposed algorithm can also be used after some small changes are made. A detailed process was illustrated by Li and He (2002).

For an edge weighted network, each edge can be in one of two states, operative or failed. Therefore, the network also has two states, operative or failed. In order to use Boolean algebra operators, the operative state and failed state are represented by 1 and 0, respectively. Two terminal nodes are denoted as source and sink. A network structure function is represented by $\Phi(G)$ and defined as follows

$$\Phi(G) = \begin{cases} 1 & \text{if network operates} \\ 0 & \text{if network fails} \end{cases} \quad (1)$$

Apparently, if all edges of any path of a network are in an operative state, the network operates. Assuming the network structure function and all the edges in a network are Boolean variants, the network structure function can be written as

$$\Phi(G) = \bigcup_{k=1}^l A_{pk} \quad (2)$$

where l is the number of paths of G , A_{pk} is the k th path of G and can be written as

$$A_{pk} = a_{k1} a_{k2} \cdots a_{ks} \quad (3)$$

where a_{ki} is an edge of G and s is the number of edges in A_{pk} .

For any path A_{pi} in Eq. (3), if it is not a minimal path (MP), there must be a corresponding MP A_{pj} , which is formed by removing several edges from A_{pi} . According to Boolean calculation, $A_{pi} \cup A_{pj} = A_{pj}$ exists. As a result, the network structure function $\Phi(G)$ can be rewritten as

$$\Phi(G) = \bigcup_{k=1}^m A_k \quad (4)$$

where m is the number of MPs of G and A_k is the k th MP of G .

For example, Fig. 1 is a bridge network with source s and sink t and $A_{p1} = acd$ is a path but not a MP. Here $A_{p2} = ad$ is its corresponding MP after removing

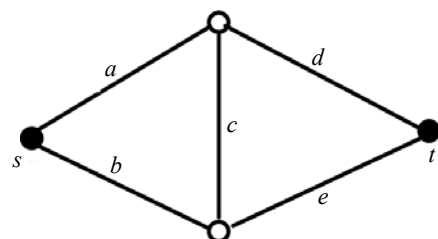


Fig. 1 A bridge network

edge c from A_{p1} . According to Boolean calculation, $A_{p1} \cup A_{p2} = acd \cup ad = ad = A_{p2}$ exists.

Also, if the MPs are disjointed, the network structure function $\Phi(G)$ can be expressed as

$$\Phi(G) = \sum_{i=1}^M L_i \quad (5)$$

where M is the number of disjointed minimal paths (DMPs) of G and L_i is the i th DMP of G .

From the above analysis, the goal of a general path-based algorithm is to change $\Phi(G)$ from $\Phi(G) = \bigcup_{k=1}^m A_k$ to

$$\Phi(G) = \sum_{i=1}^M L_i.$$

After defining the network structure function, the network reliability is

$$R = P(\Phi(G) = 1) = \sum_{i=1}^M P(L_i = 1) \quad (6)$$

Correspondingly, a network complementary structure function can be defined as follows:

$$\Phi'(G) = 1 - \Phi(G) = \begin{cases} 1 & \text{if network fails} \\ 0 & \text{if network operates} \end{cases} \quad (7)$$

As the network fails if all edges of any cut of the network are in a failure state, the complementary structure function can be expressed as

$$\Phi'(G) = \sum_{j=1}^K C_j \quad (8)$$

where K is the number of disjointed minimal cuts (DMCs) of G and C_j is the j th DMC of G .

The failure probability of the network is

$$F = P(\Phi'(G) = 1) = \sum_{j=1}^K P(C_j = 1) \quad (9)$$

For the above structure function, three theorems have been proven by authors (see appendix) and are stated below.

Theorem 1: Let G represent a connected network and a_r be an edge of G . Suppose that G_1 is a subgraph of G , which is formed by removing a_r from G . Then,

$$\bar{a}_r \Phi(G_1) = \bar{a}_r \Phi(G) \quad (10)$$

Theorem 2: Let G represent a connected network and a_r be an edge connecting the source with node J of G . Suppose that G_1 is a subgraph of G by removing a_r and assuming J as a new source. Then

$$a_r \Phi(G_1) = a_r \Phi(G) \quad (11)$$

Theorem 3: Let G represent a connected network and a_r be an edge connecting the source with node J of G . Suppose that G_1 is a subgraph of G by merging J into source. Then

$$a_r \Phi(G_1) = a_r \Phi(G) \quad (12)$$

Herein the process of merging node J into another node I is to delete node J and the corresponding edge connecting nodes I and J . In the meantime, change the edge connecting with node J to connect with node I .

3 Path-based recursive decomposition algorithm

In order to simplify the calculation, the shortest MP from the source to the sink is defined as $A_1 = a_{11}a_{12} \cdots a_{1m_1}$, where a_{1i} is an edge of the network and m_1 is the number of edges in A_1 . From A_1 , according to Boolean laws, the structure function $\Phi(G)$ in Eq. (4) can be written as

$$\Phi(G) = A_1 \bigcup_{k=1}^m A_k = A_1 \cup \Phi(G) = A_1 + \bar{A}_1 \Phi(G) \quad (13)$$

Using De Morgan's Law, $\bar{A}_1$ can be expanded as:

$$\bar{A}_1 = \bar{a}_{11} + a_{11}\bar{a}_{12} + \cdots + a_{11}a_{12} \cdots a_{1i-1}\bar{a}_{1i} + \cdots + a_{11}a_{12} \cdots \bar{a}_{1m_1} \quad (14)$$

Substituting Eq. (14) into Eq. (13) and using Theorem 1 and Theorem 3, the network structure function $\Phi(G)$ becomes

$$\begin{aligned} \Phi(G) = & A_1 + \bar{a}_{11}\Phi(G_{11}) + a_{11}\bar{a}_{12}\Phi(G_{12}) + \cdots \\ & + a_{11}a_{12} \cdots a_{1i-1}\bar{a}_{1i}\Phi(G_{1i}) + \cdots + a_{11}a_{12} \cdots \bar{a}_{1m_1}\Phi(G_{1m_1}) \end{aligned} \quad (15)$$

where G_{1i} represents a subgraph after removing edge a_{1i} from G and merging the nodes connecting with the edge a_{1j} ($j=1,2,\dots,i-1$) into the source in sequence.

Note that the original recursive decomposition algorithm (Li and He, 2002) obtains the subgraph G_{1i} only by removing the edge a_{1i} in the above equation, without utilizing information contained in the edge a_{1j} ($j=1,2,\dots,i-1$). As a result, the subgraphs obtained from the original recursive decomposition algorithm are more complicated, and consequently, more calculated disjoint events and more computational efforts are needed.

The subgraphs can be classified into connected subgraphs and disconnected subgraphs. If the number of connected subgraphs is taken as m_{1p} , then the number of disconnected subgraphs is $m_{1c} = m_1 - m_{1p}$. In the meantime, if G_{1i} is a disconnected subgraph, the coefficients $a_{11}a_{12} \cdots a_{1i-1}\bar{a}_{1i}$, in front of $\Phi(G_{1i})$ can be equal to C_{1j} . As G_{1i} is a disconnected subgraph after

removing the edge a_{li} from the connected network G , according to the definition of minimal cut (MC), $\bar{a}_{li}$ is a MC as well. According to the process of expanding A_i in Eq. (14), C_{lj} is a DMC of G . Then, the complementary structure function can be written as

$$\Phi'(G) = \sum_{j=1}^{m_{lc}} C_{lj} + Q_1 \quad (16)$$

where Q_1 is the remainder term, denoting that C_{lj} ($j=1,2,\dots, m_{lc}$) are still not the entire DMC set of the network.

As no path exists in the disconnected subgraphs, the structure functions of the disconnected subgraphs are equal to 0, and $\Phi(G)$ can be rewritten as

$$\Phi(G) = A_1 + \sum_{i=1}^{m_{lp}} B_{li} \Phi(G_{li}) \quad (17)$$

where $B_{li} = a_{11}a_{12}\dots a_{li-1}\bar{a}_{li}$ and G_{li} denote the first order connected subgraph.

The connected subgraphs can be further decomposed into second order subgraphs using their shortest MPs. The corresponding coefficients in front of the structure function of a second order disconnected subgraph also form a DMC of G . Thus, the expression of $\Phi'(G)$ can be further written as

$$\Phi'(G) = \sum_{j=1}^{m_{lc}} C_{lj} + \sum_{j=1}^{m_{2c}} C_{2lj} + Q_2 \quad (18)$$

where Q_2 is also the remainder term, but $P(Q_2 = 1) < P(Q_1 = 1)$.

$\Phi(G)$ can be written as

$$\Phi(G) = A_1 + \sum_{i=1}^{m_{lp}} B_{li} A_{2i} + \sum_{i=1}^{m_{lp}} \sum_{l=1}^{n_{2ip}} B_{li} B_{2il} \Phi(G_{2il}) \quad (19)$$

where A_{2i} is a shortest MP of G_{li} , B_{2il} is the coefficient by decomposing $\bar{A}_{2i}$ according to Eq. (14), n_{2ip} is the number of the connected subgraphs by decomposing G_{li} and G_{2il} denotes the second order connected subgraph.

The above connected subgraphs are decomposed step by step until no connected subgraph exists, then all the DMPs will be enumerated. Finally $\Phi(G)$ can be expressed as

$$\Phi(G) = A_1 + \sum_{i=1}^{m_{lp}} B_{li} A_{2i} + \sum_{i=m_{lp}+1}^{m_{2p}} B_{li} A_{3i} + \dots + \sum_{i=m_{(n-1)p}+1}^{m_{np}} B_{li} A_{(n+1)i} \quad (20)$$

where B_i is a coefficient by expanding MPs, A_{ki} is a shortest MP of a connected subgraph and m_{np+1} is the number of all connected graphs.

Let $L_1 = A_1$, $L_{i+1} = B_i A_{ki}$ ($k=1,2,\dots, n+1$; $i=1,2,\dots, m_{np}$), then Eq. (20) can be transformed into:

$$\Phi(G) = \sum_{i=1}^M L_i \quad (21)$$

where L_i is a DMP and $M = m_{np} + 1$.

At the same time, all DMCs are also enumerated, that is

$$\Phi'(G) = \sum_{j=1}^K C_j \quad (22)$$

where C_j is a DMC and K is the number of all DMCs.

After enumerating all DMPs and DMCs, the network reliability can be deduced

$$R = P[\Phi(G) = 1] = \sum_{i=1}^M P(L_i = 1) \quad (23)$$

and the network failure probability is

$$F = P[\Phi(G) = 0] = \sum_{j=1}^K P(C_j = 1) \quad (24)$$

For a large network, it might be impossible to enumerate all disjoint products. In these cases, the upper and lower bounds of the network reliability can be calculated from the probabilistic inequality (Li, 2005)

$$\sum_{i=1}^{M_s} P(L_i = 1) \leq R \leq 1 - \sum_{j=1}^{K_f} P(C_j = 1) \quad (25)$$

where M_s and K_f are the numbers of the calculated DMPs and DMCs, respectively, $M_s \leq M$ and $K_f \leq K$.

When the difference between the upper bound and the lower bound is less than a specified error bound, the network reliability can be given as

$$R = 0.5 \left(1 - \sum_{j=1}^{K_f} P(C_j = 1) + \sum_{i=1}^{M_s} P(L_i = 1) \right) \quad (26)$$

4 Technology to accelerate the recursive decomposition algorithm

If the edge failures are mutually s -independent, two network reduction techniques, i.e., series-edge reduction technology and parallel-edge reduction technology (Page and Perry, 1988; 1989) can be used to simplify the network and speed up the path-based recursive decomposition algorithm.

A general series-edge reduction technology is illustrated in Fig. 2. When a node has just two incident edges, if it is not source or a sink, it is deleted and the two incident edges are replaced by an edge whose reliability is equal to the product of the two incident

edge reliabilities. For example, in Fig. 2, the node I is deleted and $p_e = p_a p_b$.

A general parallel-edge reduction technology is illustrated in Fig. 3. When two or more edges have the same incident nodes, these edges can be replaced by an edge whose failure probability is equal to the product of the failure probabilities for these edges. For example, in Fig. 3, $q_e = q_a q_b$, that is $p_e = 1 - (1 - p_a)(1 - p_b)$.

In fact, these two network reduction techniques can

be used repeatedly until the network cannot be simplified further. For example, for a network with six edges as indicated in Fig. 4, after using series-edge reduction and parallel-edge reduction repeatedly, it can be represented by a network with only one edge h while $p_h = 1 - \{1 - [1 - (1 - p_a p_b)(1 - p_c)]p_f\}(1 - p_d p_e)$.

Incorporating the above network reduction techniques, the proposed algorithm is designed as follows:

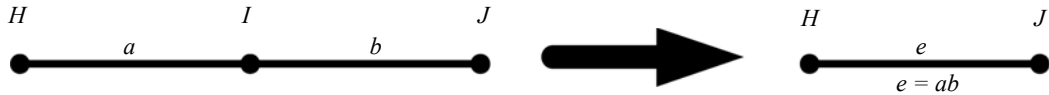


Fig. 2 Series-edge reduction



Fig. 3 Parallel-edge reduction

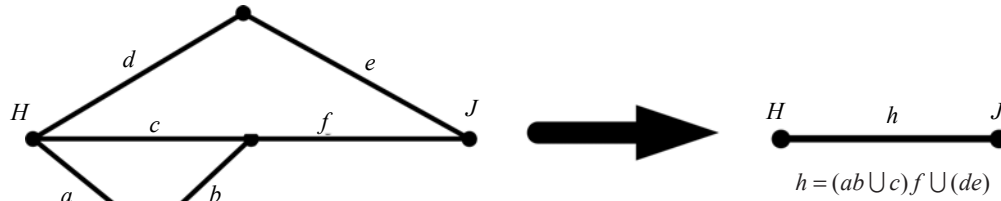


Fig. 4 A simple example network

(1) Use network reduction techniques to simplify the network and search one shortest MP of the network by using the breadth first searching algorithm (BFS) (Li and He, 2002);

(2) A DMP is generated by combining the shortest MP with the coefficients in front of the network structure function. The probability of this DMP is accumulated into the lower bound. Use Eqs.(13)–(15) to decompose the network into subgraphs;

(3) For the disconnected graphs among the above subgraphs, the coefficients in front of their structure functions form the DMCs. The probability of these DMCs is subtracted from the upper bound. Meanwhile, these disconnected subgraphs are eliminated. For the connected subgraphs, network reduction techniques are used to search for the shortest MP using BFS;

(4) Repeat steps (2) and (3) until no connected subgraphs exist or the difference between the upper bound and lower bound of reliability satisfies the specified error bound.

(5) Calculate the reliability of the network using Eq. (26).

In Fig. 5, a simple bridge network shown in Fig. 1 is used to illustrate the process of the proposed algorithm and three DMPs and two DMCs are generated.

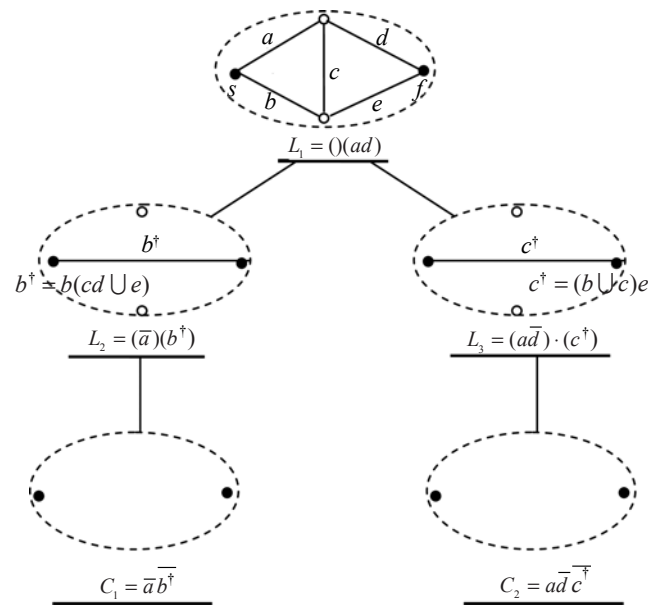


Fig. 5 Process of the proposed algorithm to decompose a bridge network (s represents the source and t represents the sink) (For any DMP, the value in the first bracket is the coefficient in front of current network structure function while the coefficient in the second bracket represents a MP of current network)

5 Case studies

A Pentium IV 3.0 GHz personal computer with 1 G RAM is used to execute the computation in the following examples.

Example 1

Figure 6 shows a network with 17 nodes and 32 edges. The edges failures are mutually s -independent. The source is node s and the sink is node t . Let all edge

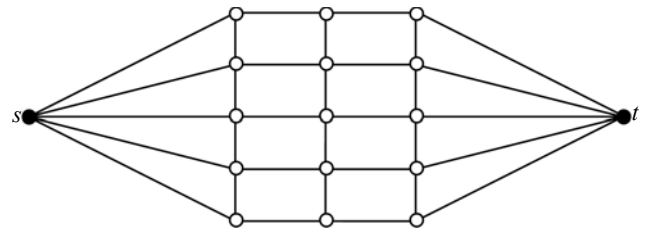


Fig. 6 A network with 17 nodes and 32 edges

Table 1 Calculated results for Case I

Error bound (10^{-3})	Reliability			Computed event number		Time (s)
	Lower bound	Upper bound	Solution	DMP	DMC	
0		0.999930		17,523	15,637	1.92
0.1	0.999851	0.999950	0.999901	954	48	0.34
0.2	0.999761	0.999961	0.999861	537	14	0.31
0.5	0.999479	0.999978	0.999728	355	5	0.28
1.0	0.998993	0.999990	0.999492	193	1	0.27

Table 2 Calculated results for Case II

Error bound (10^{-3})	Reliability			Computed event number		Time (s)
	Lower bound	Upper bound	Solution	DMP	DMC	
0		0.996522		17,523	15,637	1.92
0.1	0.996486	0.996586	0.996536	8,451	3,072	1.11
0.2	0.996447	0.996647	0.996547	6,406	1,712	0.90
0.5	0.996298	0.996798	0.996548	4,661	1,008	0.73
1.0	0.996029	0.997029	0.996529	3,155	464	0.57

Table 3 Calculated results for Case III

Error bound (10^{-3})	Reliability			Computed event number		Time (s)
	Lower bound	Upper bound	Solution	DMP	DMC	
0		0.017194		17,523	15,637	1.92
0.1	0.017194	0.017294	0.017244	17,246	14,180	1.86
0.2	0.017194	0.017393	0.017293	17,126	13,705	1.84
0.5	0.017193	0.017682	0.017438	16,299	11,770	1.77
1.0	0.017191	0.018168	0.017680	16,152	11,415	1.75

Table 4 Calculated results for Case IV

Error bound (10^{-3})	Reliability			Computed event number		Time (s)
	Lower bound	Upper bound	Solution	DMP	DMC	
0		0.000777		17,523	15,637	1.92
0.1	0.000777	0.000877	0.000827	14,786	9,412	1.62
0.2	0.000777	0.000974	0.000876	13,749	7,683	1.53
0.5	0.000777	0.001277	0.001027	12,892	6,771	1.47
1.0	0.000777	0.001767	0.001272	10,140	4,092	1.23

reliabilities be 0.9 (Case I), 0.8 (Case II), 0.2 (Case III) and 0.1 (Case IV), respectively. Considering different error bounds, the analysis results using the proposed algorithm are shown in Tables 1–4. Figure 7 shows the related curves between the error bound and computation time for four cases using the proposed algorithm.

From Tables 1–4 and Fig. 7, note that the computation time and the number of decomposed disjoint events increase as the error bound decreases. When all edges own high reliabilities (Case I and Case II) and the error bound is between 0.001 and 0.0001, the computation time and the number of decomposed disjoint events of approximate solution are much less than for an accurate solution. For example, when all edge reliabilities are 0.9 and the error bound is 0.0001, the computation time is only 0.34 s, which is only 17.8% of the accurate solution. However, when all edges own low reliabilities (Case III and Case IV) and the error bound is between 0.001 and 0.0001, the computation time and the number of decomposed disjoint events of the approximate solution is almost the same as those for the accurate solution. For example, when all edge reliabilities are 0.2 and the error bound is 0.001, the computation time is 1.75 s, i.e., 91.1% of the accurate solution. Therefore, the proposed algorithm may be more suitable for calculating the approximate reliability of a network with higher edge reliability.

In addition, to compare the efficiency between the original recursive decomposition algorithm (Li and

He, 2002) and the algorithm proposed in this paper, the accurate solutions of four cases are evaluated using the original recursive decomposition algorithm. The results are shown in Table 5. Comparing the results in Table 5 and those in Tables 1–4, it is seen that the efficiency of the original recursive decomposition algorithm is much lower than the proposed algorithm, though more time is spent on executing network reduction program in the proposed algorithm. In fact, in order to obtain the same results, 1.16 million DMPs and 2.22 million DMCs, which are larger than those in the proposed algorithm by 66 times and 142 times, respectively, are run in the original recursive decomposition algorithm. As for the computation time, the original algorithm takes about 35 times longer.

Example 2

Figure 8 shows an actual water distribution network located in a small city in China, with 1 water factory, 44 pipelines and 30 consumers. This water distribution network is considered to be a network with 1 source, 44 edges and 30 sinks. In Fig. 8, node 1 is the source and nodes 2–31 are sinks. Edge reliabilities, shown in Table 6, are evaluated using the pipeline seismic reliability analysis method (Liu *et al.*, 2005), assuming an earthquake intensity of VII. Applying the proposed algorithm, the seismic reliability of the water distribution network from the water factory to each consumer was evaluated in much less computation time, and the results are shown in Table 7.

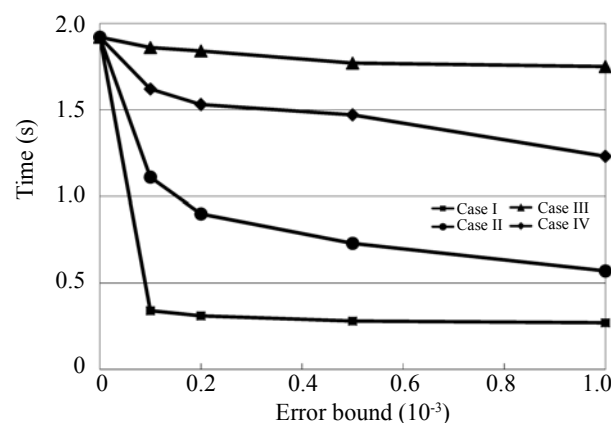


Fig. 7 Computation time for four case studies using the proposed algorithm

Table 5 Calculated results using the original recursive decomposition algorithm

Case	Reliability	Computed event number		Time (s)
		DMP	DMC	
I	0.999930	1,162,914	2,222,958	68
II	0.996522			
III	0.017194			
IV	0.000777			

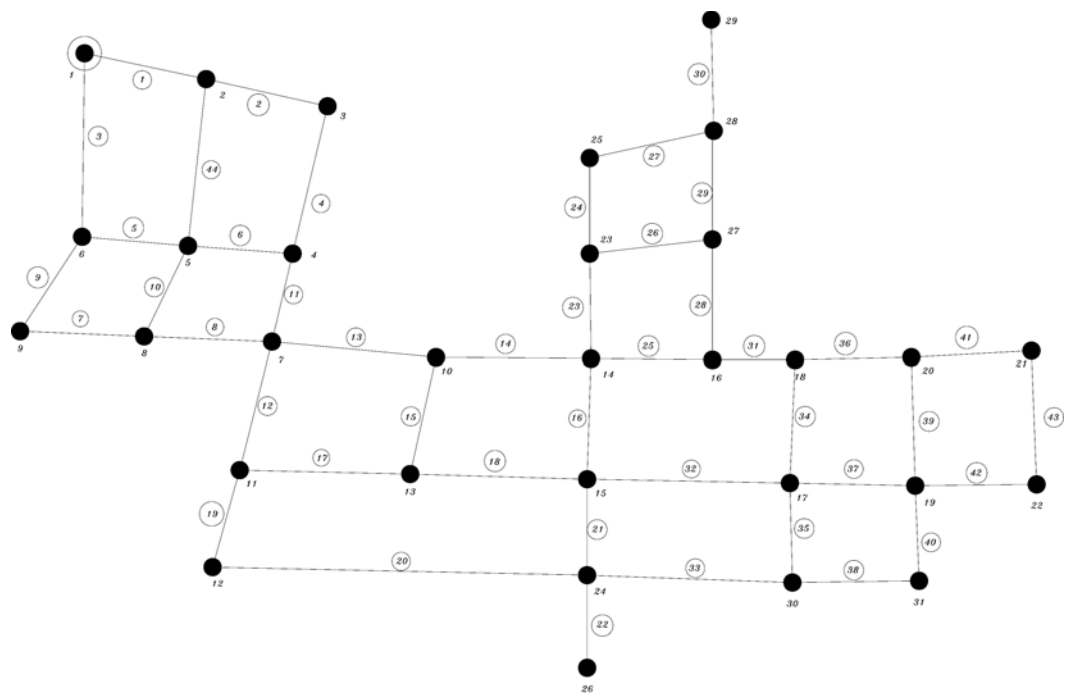


Fig. 8 A water distribution network

Table 6 Pipelines reliabilities of the water distribution network

Edge No.	Reliability	Edge No.	Reliability
1	0.9132	23	0.5328
2	0.9652	24	0.5452
3	0.8772	25	0.5286
4	0.9015	26	0.5481
5	0.5897	27	0.4981
6	0.5573	28	0.5129
7	0.4238	29	0.5238
8	0.4878	30	0.4819
9	0.8135	31	0.5128
10	0.8754	32	0.6192
11	0.9325	33	0.6253
12	0.7498	34	0.5677
13	0.9154	35	0.5368
14	0.5872	36	0.5486
15	0.6752	37	0.5238
16	0.8573	38	0.5513
17	0.7695	39	0.5312
18	0.6892	40	0.5126
19	0.7352	41	0.5337
20	0.6685	42	0.5286
21	0.7682	43	0.5326
22	0.7548	44	0.9138

Example 3

Figure 9 shows an actual gas supply network located in a city of China, with 2 gas factories, 492 pipelines and 391 nodes. For this large-scale network, the pipeline is viewed as the edges and the gas factories are viewed as sources. Edge reliabilities are also evaluated using the pipeline seismic reliability analysis method (Liu *et al.*, 2005) assuming that the earthquake intensity is VII. Using a virtual source which connects each gas factory to an edge with a reliability of 1, the network becomes a single-source network. The proposed algorithm is employed and the error bound is set at 0.001. For this large network, the seismic reliability from the sources to the main terminals can be readily calculated and partial results are shown in Table 8.

6 Conclusion

The analytical method presented in the paper is, in fact, a path-based recursive decomposition algorithm for evaluating network reliability. For small and medium lifeline networks, the proposed algorithm can derive an accurate reliability by calculating all DMPs and DMCs. For large lifeline networks, the proposed algorithm provides a tool to obtain an approximate reliability assessment, which satisfies a prescribed error bound. The case studies presented in this paper indicate that the proposed algorithm can be used to evaluate the seismic reliability of lifeline systems, including water distribution networks, gas supply networks, etc. Compared to the original recursive decomposition algorithm, the proposed algorithm provides much more efficient results. The case studies of a network with

Table 7 Evaluation results of the water distribution network

Terminal No.	Reliability	Computation time(s)	Terminal No.	Reliability	Computation time(s)
2	0.9665	0.06	17	0.6074	0.84
3	0.9548	0.05	18	0.4944	0.98
4	0.9319	0.05	19	0.4530	1.08
5	0.9573	0.06	20	0.3958	1.65
6	0.9556	0.06	21	0.2777	2.88
7	0.8995	0.06	22	0.2935	2.65
8	0.9144	0.06	23	0.4662	1.01
9	0.8495	0.06	24	0.7055	0.41
10	0.8630	0.35	25	0.3042	1.54
11	0.8176	0.34	26	0.5325	0.45
12	0.7083	0.61	27	0.4098	1.18
13	0.8083	0.41	28	0.2820	0.96
14	0.7407	0.49	29	0.1359	0.96
15	0.7564	0.46	30	0.5723	1.12
16	0.5283	1.23	31	0.4134	1.64



Fig. 9 A gas supply network

Table 8 Evaluation results of the gas network

Terminal No.	Reliability	Computation time (s)
1	0.93425	15
45	0.99718	7
68	0.99392	5
313	0.99388	1
362	0.95486	4

different edge reliabilities indicate that the proposed algorithm may be more suitable for calculation of an approximate reliability of lifeline networks with high edge reliabilities.

Acknowledgements

The support of the Natural Science Funds for Innovative Research Groups of China (Grant No.50621062) is greatly appreciated.

References

- Abraham JA (1979), "An Improved Algorithm for Network Reliability," *IEEE Transactions on Reliability*, **28**(1): 58–61.
- Aggarwal KK and Misra KB (1975), "A Fast Algorithm for Reliability Evaluation," *IEEE Transactions on Reliability*, **24**(1): 83–85.
- Dotson WP and Gobien JO (1979), "A New Analysis Technique for Probability Graphs," *IEEE Transactions on Circuits & Systems*, **26**: 855–865.
- Fratta L and Montanari UG (1973), "A Boolean Algebra Method for Computing the Terminal Reliability in Communication Network," *IEEE Transactions on Circuit Theory*, **20**(3): 203–211.
- Li J (2005), *Lifeline Earthquake Engineering-basic Methods and Applications*, Beijing: Science Press. (in Chinese)
- Li J and He J (2002), "A Recursive Decomposition Algorithm for Network Seismic Reliability Evaluation," *Earthquake Engineering and Structural Dynamics*, **31**: 1525–1539.
- Liao JS (1982a), "Sum of Disjoint Algorithm for Network Reliability Evaluation (I)," *Universe Transactions*, **3**: 51–56. (in Chinese)
- Liao JS (1982b), "Sum of Disjoint Algorithm for Network Reliability Evaluation (II)," *Universe Transactions*, **4**: 28–35. (in Chinese)
- Liu W, Li J, Wei SI and Zhou Jh (2005), "Reliability Analysis of Gas Pipeline Under Earthquake," *Earthquake Resistant Engineering and Retrofitting*, **27**(6): 94–96. (in Chinese)
- Page LB and Perry JE (1988), "A Practical Implementation of the Factoring Theorem for Network Reliability," *IEEE Transactions on Reliability*, **37**(3): 259–267.
- Page LB and Perry JE (1989), "Reliability of Directed Networks Using the Factoring Theorem," *IEEE Transactions on Reliability*, **38**(4): 556–562.
- Yoo YB and Deo N (1988), "A Comparison of Algorithm for Terminal-pair Reliability," *IEEE Transactions on Reliability*, **37**(2): 210–215.

Appendix

Definition: Ω_G represents a MP set including all MPs of network G and element 0. In other word, $\Omega_G = \{0, A_1, A_2, \dots, A_m\}$, where A_k ($k=1, 2, \dots, m$) is a MP of network G .

Apparently, the network structure function $\Phi(G)$ is an union of all elements in the set Ω_G . In order to prove following theorems conveniently, let $a\Omega_G = \{0, aA_1, aA_2, \dots, aA_m\}$, where a is an edge of the network G .

Theorem 1: Let G represent a connected network and a_f be an edge of G . Suppose that G_1 is a subgraph of G , which is formed by removing a_f from G . Then,

$$\bar{a}_f \Phi(G_1) = \bar{a}_f \Phi(G) \quad (A1)$$

Proof: According to Eq.(4), the right of the Eq. (A1) can be written as

$$\bar{a}_f \Phi(G) = \bigcup_{k=1}^m (\bar{a}_f A_k) \quad (A2)$$

and the left the Eq. (A1) can be written as

$$\bar{a}_f \Phi(G_1) = \bigcup_{k=1}^n (\bar{a}_f A_{1k}) \quad (A3)$$

where A_k and A_{1k} represent MPs of network G and G_1 , respectively.

Apparently, if the equation $\bar{a}_f \Omega_G = \bar{a}_f \Omega_{G_1}$ can be proved, the theorem is proved.

First considering the set $\bar{a}_f \Omega_G$, the MPs in the set Ω_G can be classified into two cases.

- (1) It consists of edge a_f or
- (2) it doesn't consist of edge a_f .

For case (1), any element, $\bar{a}_f A_k$, equals 0 because the intersection of $\bar{a}_f$ and a_f equals 0. For case (2), A_k must be a MP of graph G_1 according to the relationship between G_1 and G . From the above analysis, it can be deduced that $\bar{a}_f A_k \in \bar{a}_f \Omega_{G_1}$. Therefore

$$\bar{a}_f \Omega_G \subset \bar{a}_f \Omega_{G_1} \quad (A4)$$

Considering the set $\bar{a}_f \Omega_{G_1}$, if subgraph G_1 is a disconnected graph, then $m=0$ and $\bar{a}_f \Phi(G_1) = \{0\}$. If the graph G_1 is a connected subgraph, A_{1k} must be a MP of graph G which does not consist of edge a_f . So it can be deduced that $\bar{a}_f A_{1k} \in \bar{a}_f \Omega_G$. Therefore

$$\bar{a}_f \Omega_G \supset \bar{a}_f \Omega_{G_1} \quad (A5)$$

From Eq.(A4) and Eq.(A5), there exists

$$\bar{a}_f \Omega_G = \bar{a}_f \Omega_{G_1} \quad (A6)$$

The theorem is proved.

Theorem 2: Let G represent a connected network and a_f be an edge connecting the source with node J of G . Suppose that G_1 is a subgraph of G by removing a_f and assuming J as a new source. Then

$$a_f \Phi(G_1) = a_f \Phi(G) \quad (A7)$$

Proof: According to Eq. (4), the right of the Eq. (A7) can be written as

$$a_r \Phi(G) = \bigcup_{k=1}^m (a_r A_k) \quad (\text{A8})$$

and the left the Eq. (A7) can be written as

$$a_r \Phi(G_1) = \bigcup_{k=1}^n (a_r A_{1k}) \quad (\text{A9})$$

where A_k and A_{1k} represent MPs of network G and G_1 , respectively.

Apparently, if the equation $a_r \Omega_G = a_r \Omega_{G_1}$ can be proved, the theorem is proved.

First considering the set $a_r \Omega_G$, the MPs in the set Ω_G can be classified into two cases.

- (1) It consists of edge a_r or
- (2) it doesn't consist of edge a_r .

For case (1), any element, $a_r A_k$, can be written as

$$a_r A_k = a_r A_k' \quad (\text{A10})$$

where A_k' consists all edges of A_k except a_r . Apparently, according to the process of generating G_1 , A_k' must be a MP of G_1 from its new source H . For case (2), A_k must be a MP of graph G_1 according to the relationship between G_1 and G . Therefore, it can be deduced that $a_r A_k \in a_r \Omega_{G_1}$. There must be

$$a_r \Omega_G \subset a_r \Omega_{G_1} \quad (\text{A11})$$

For the set $a_r \Omega_{G_1}$, the MPs in the set Ω_{G_1} can be classified into two cases as well.

- (1) Its start node is the new source H , or
- (2) its start node is an original source.

For case (1), $a_r A_{1k}$ can be written as

$$a_r A_{1k} = a_r A_{1k}' \quad (\text{A12})$$

where A_{1k}' is formed by adding edge a_r to A_{1k} . Apparently, according to the process of generating G_1 , A_{1k}' must be a MP of graph G . For case (2), A_{1k} must be a MP of graph G according to the relationship between G_1 and G . Therefore, it can be deduced that $a_r A_{1k} \in a_r \Omega_G$. There must be

$$a_r \Omega_G \supset a_r \Omega_{G_1} \quad (\text{A13})$$

From Eq. (A11) and Eq. (A13), there exists

$$a_r \Omega_G = a_r \Omega_{G_1} \quad (\text{A14})$$

The theorem is proved.

Theorem 3: Let G represent a connected network and a_r be an edge connecting the source with node J of G . Suppose that G_1 is a subgraph of G by merging J into source. Then

$$a_r \Phi(G_1) = a_r \Phi(G) \quad (\text{A15})$$

The theorem can be easily concluded from theorem 2.